DP-S4S: Accurate and Scalable Select-Join-Aggregate Query Processing with User-Level Differential Privacy

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Differential Privacy (DP)

Dataset	Privacy Notion	Neighboring Dataset

Dataset

User	Age
Alice	25
Bob	45

Record-level DP

Neighboring Dataset

User	Age
Alice	25
Bob	45
Cathy	**60**

Dataset

Customer	Order
Alice	**#1**
Alice	**#2**
Bob	#3

User-level DP w/o self-join

Neighboring Dataset

User	Order
Bob	#3

Dataset

Customer	Order	Supplier
Alice	**#1**	**Bob**
Alice	#2	Cathy
Bob	**#3**	Cathy

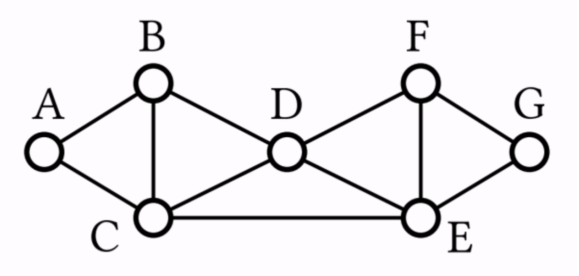
User-level DP w/ self-join

Neighboring Dataset

Customer	Order	Supplier
Alice	#2	Cathy

Select-Join-Aggregate (SJA) Queries

```
SELECT count(*)
FROM edge as e1, edge as e2, edge as e3
WHERE e1.dst = e2.src AND e2.dst = e3.src
    AND e3.dst = e1.src AND e1.src < e2.src
    AND e2.src < e3.src
```



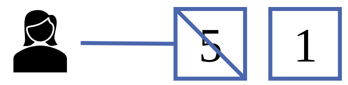
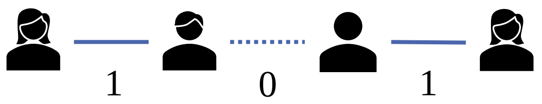
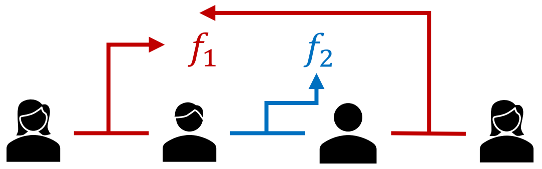
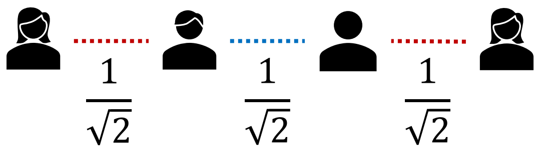
$$Q(\mathrm{I}) \coloneqq f(T_Q) = \sum_{t \in T} w_t$$

Database Instance

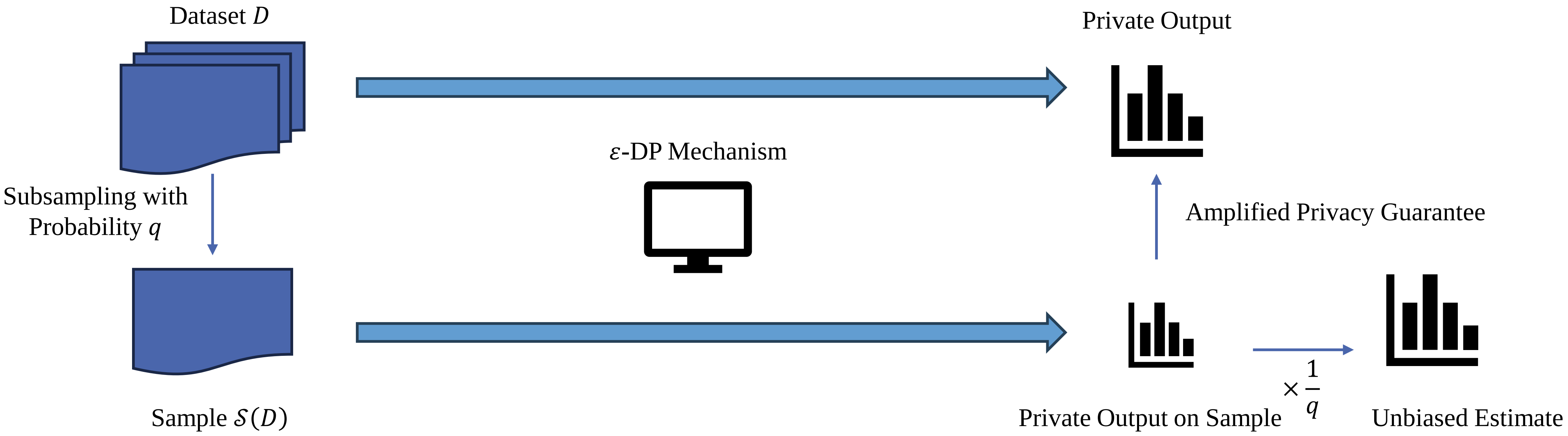
Flat Table of Join Results

Weight of tuple t

SJA Queries under DP: Truncation

Privacy Notion		Truncated Query	Example ($\tau = 1$)	Ref.
Record-level DP		$\bar{f}(T) \coloneqq \sum_t \min\{w_t, \tau\}$	 (truncate 5 to 1)	Strawman, e.g. [ACG+, CCS'16] [QDY+, KDD'22]
User-level DP w/ self-join	Scalar query (LP)	$\bar{f}(T) \coloneqq \max_{x_t} \sum_t x_t$ $s.t.\ \ \sum_{t:u \rightsquigarrow t} x_t \le \tau$ for any u; $0 \le x_t \le w_t$ for any t	 (truncate the middle edge)	[KNR+, TCC'13] [DFY+, SIGMOD'22]
	Vector query (QCQP)	$\bar{f}(T) \coloneqq \sum_t w_{f,t} \cdot z^*_{f,t}$, where z^* is the optimal solution to $\max_{y_u, z_{f,t}} \sum_u y_u$ $s.t.\ \sum_f \left(\sum_{t:u \rightsquigarrow t} w_{f,t} \cdot z_{f,t}\right)^2 \le \tau^2$ for any u; $\sum_{u:u \rightsquigarrow t}(y_u - 1) \le z_{f,t} - 1$ for any f, t; $0 \le y_u, z_{f,t} \le 1$ for any u, f, t	 (truncate weights) 	[DSY, SIGMOD'23]

Privacy Amplification by Subsampling



Record-level DP

If $\mathcal{M}$ is ε-DP, then $\mathcal{M} \circ \mathcal{S}$ is $\varepsilon' = \ln\left(1 + q(e^{\varepsilon} - 1)\right)$-DP

[BBG, NeurIPS’18]

$\approx q\varepsilon$

Our Contribution (1): Sample-and-Truncate Amplification

- S&E [FY, SIGMOD'24]
  - Sample a user, and Explore its collaborators
  - Each t is sampled with prob $\frac{1}{|U|}$
  - C: the upper bound on the number of collaborators per user
  - Samples are identical if the *witness* user is not in the sample, which happen with prob $1 - \frac{C}{|U|}$
  - $\varepsilon' = \ln\left(1 + \frac{|U|}{C}(e^{\varepsilon} - 1)\right)$

- DP-S4S (This work)
  - Sample an aggregation unit
  - Each t is sampled i.i.d. with prob q
  - Δ: the upper bound on the number of aggregation units per user
  - Samples are identical with probability $p_0 = (1-q)^{\Delta}$
  - Samples differ by 1 record with probability $p_1 = \binom{\Delta}{1} q(1-q)^{\Delta}$
  - …
  - $\varepsilon' = \ln\left(\sum_{k=0}^{\Delta} p_k\, e^{\min\{k,\tau\}\varepsilon/\tau}\right)$

Our Contribution (2): Smooth Sensitivity under Rényi-DP

- SS [NRS, STOC'07]
  - S is a γ-smooth bound for LS_f if
    - $S(D) \geq LS_f(D)$ for all D
    - $S(D) \leq e^{\gamma} \cdot S(D')$ for all $D \sim D'$
  - The smooth sensitivity

$$SS(D,\gamma) \coloneqq \max_{D^*} LS_f(D^*) \cdot e^{-\gamma\, d(D,D^*)}$$

is a γ-smooth bound for LS_f

  - $\mathcal{M}_{SS}(D) \coloneqq f(D) + \mathcal{N}(0,\sigma^2)$ satisfies (ε,δ)-DP, for

$$\sigma = SS(D,\gamma) \cdot \frac{5\sqrt{2\ln\frac{2}{\delta}}}{\varepsilon},$$

where $\gamma = \frac{\varepsilon}{4\left(d+\ln\frac{2}{\delta}\right)}$.

- RDP-SS (This work)
  - $\mathcal{M}_{RDP-SS}(D) \coloneqq f(D) + \mathcal{N}(0,\sigma^2)$ satisfies (α,ρ)-RDP, for

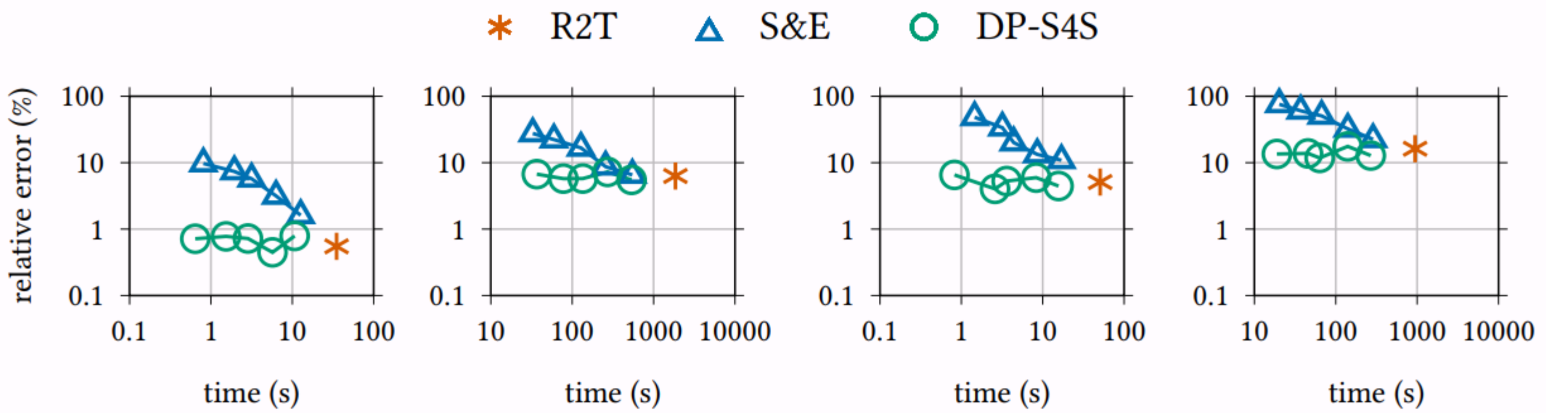
$$\sigma = SS(D,\gamma) \cdot \frac{\sqrt{\alpha/\rho}}{1-\alpha+\alpha e^{-2\gamma}},$$

where $\gamma = \frac{1}{2}\min\{t_1^{-1}, t_2\}$,

where $0 < t_1 < 1 < t_2$ are solutions to

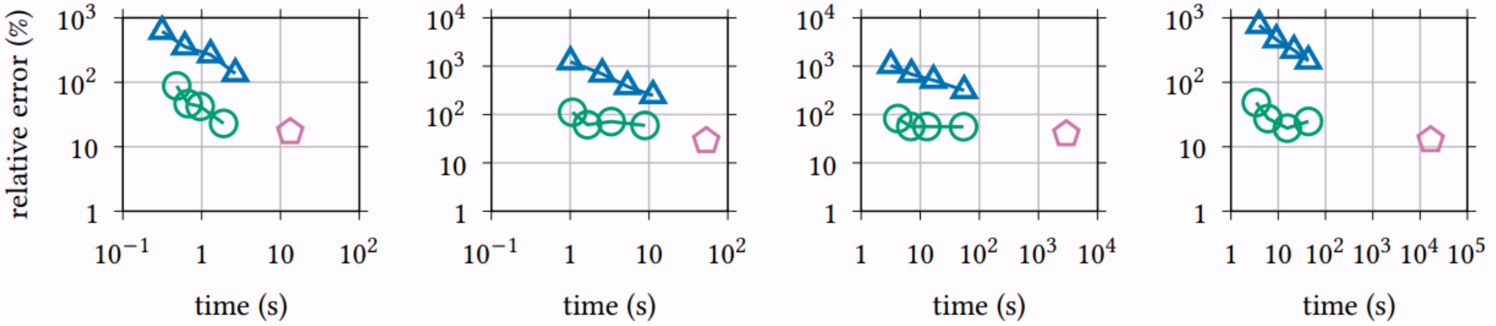
$$t^{\alpha} - \alpha e^{\frac{(\alpha-1)\rho}{d}} t + (\alpha-1)e^{\frac{(\alpha-1)\rho}{d}} = 0$$

Summary



(a) Q_{1-} (base query time: 0.18s). (b) Q_{2-} (base query time: 57.4s). (c) $Q_{\triangle}$ (base query time: 8.87s). (d) $Q_{\square}$ (base query time: 296s).

Fig. 2. Query error vs. time on Deezer with $\varepsilon = 1$. (Sampling rates $q \in \{\frac{1}{64}, \frac{1}{32}, \frac{1}{16}, \frac{1}{8}, \frac{1}{4}\}$)



(a) Q_{1-} (base query time: 1.46s). (b) Q_{2-} (base query time: 1.65s). (c) Q_{3-} (base query time: 1.91s). (d) $Q_{<}$ (base query time: 1.28s).

Fig. 5. Query error vs. time on C2A with $\varepsilon = 4$, $\delta = 10^{-7}$. (Sampling rates $q \in \{\frac{1}{32}, \frac{1}{16}, \frac{1}{8}, \frac{1}{4}\}$)

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